No.	Signature	Interpretation	Attack Labels	Individual Precision
1	Connection count: ⟨229.0, 251.0⟩ ∧ SYN error rate: ⟨1.0, 1.0⟩ ∧ Source bytes: ⟨0.0, 0.0⟩	High-volume payload-free SYN failures. Within two seconds, 229–251 connections reach the same host, all with SYN errors and zero source bytes. This strongly indicates Neptune-style SYN flooding. Operators can confirm it using the SYN-to-ACK ratio, target concentration, and half-open connection count.	neptune	100%
2	Connection duration: ⟨0.0, 0.0⟩ ∧ Same service rate: ⟨0.03, 0.03⟩ ∧ Service SYN error rate: ⟨1.0, 1.0⟩	Rapid multi-service SYN failures. Zero-duration connections span multiple services, and all recent attempts to the same service show SYN errors. This pattern is shared by Portsweep, Satan, and Neptune. Operators can distinguish them using destination-port fan-out, target concentration, and half-open connection counts.	neptune, portsweep, satan	100%
3	Destination bytes: ⟨0.0, 0.0⟩ ∧ Service: ⟨ecr_i, ecr_i⟩ ∧ Wrong fragment count: ⟨0.0, 0.0⟩	ICMP echo-related traffic without return data. The flow uses the `ecr_i` service, carries no destination bytes, and has no malformed fragments. This pattern occurs in Smurf flooding and reconnaissance but does not distinguish them. Operators can check ICMP type, packet rate, broadcast use, spoofing, and host fan-out.	ipsweep, nmap, portsweep, satan, smurf	95.65%
4	Destination-host reject error rate: ⟨0.0, 0.0⟩ ∧ Destination-host service SYN error rate: ⟨1.0, 1.0⟩ ∧ Wrong fragment count: ⟨0.0, 0.0⟩ ∧ Logged-in status: ⟨0, 0⟩	Repeated service-level SYN failures without successful login. Connections to the same host and service show SYN errors, with no reject errors or malformed fragments. This indicates connection-establishment failure but does not uniquely identify the listed attacks. Operators should inspect TCP flags, handshake state, target service, authentication attempts, and endpoint equality.	guess_-passwd, imap, land, neptune, nmap	99.96%
5	Destination-host different service rate: ⟨0.0, 0.0⟩ ∧ Destination-host service reject error rate: ⟨0.0, 0.0⟩ ∧ Service different host rate: ⟨1.0, 1.0⟩ ∧ Destination bytes: ⟨0.0, 0.0⟩	Same-service activity across multiple hosts without return data. Connections to the current host remain on one service, while the same service is contacted across different hosts. This suggests service-specific host sweeping but does not alone establish FTP, warez, or Land behavior. Operators should check host fan-out, service, transfer direction, FTP commands, and endpoint equality.	ftp_write, ipsweep, land, nmap, warezclient	99.28%

Table 8: Five representative signatures from `NSL-KDD`. Those signatures are used to detect denial-of-service attacks, which accounts for most of the attacks in the dataset.